S30: Do men separate women's sexual interest from friendliness less well than women do?

Question. Are men less able than women to tell a woman's friendliness apart from her sexual interest in photographs? Tested as the Gender × stimulus-category (friendly vs sexually interested) interaction on the per-category discrimination score ln(α); the claim predicts that men's deficit is larger for sexually interested than for friendly images.

Data and sample. Replication data file `d02.sav`: 144 heterosexual undergraduates (65 men, 79 women), each classifying 280 photographs into four categories. Discrimination scores `Sens_Friend` and `Sens_SI` were used as precomputed (the file holds only per-category hit/FA/miss totals, not the full confusion matrix, so they cannot be recomputed). **Exclusions: none** (0 rows missing Gender or either score); N = 144.

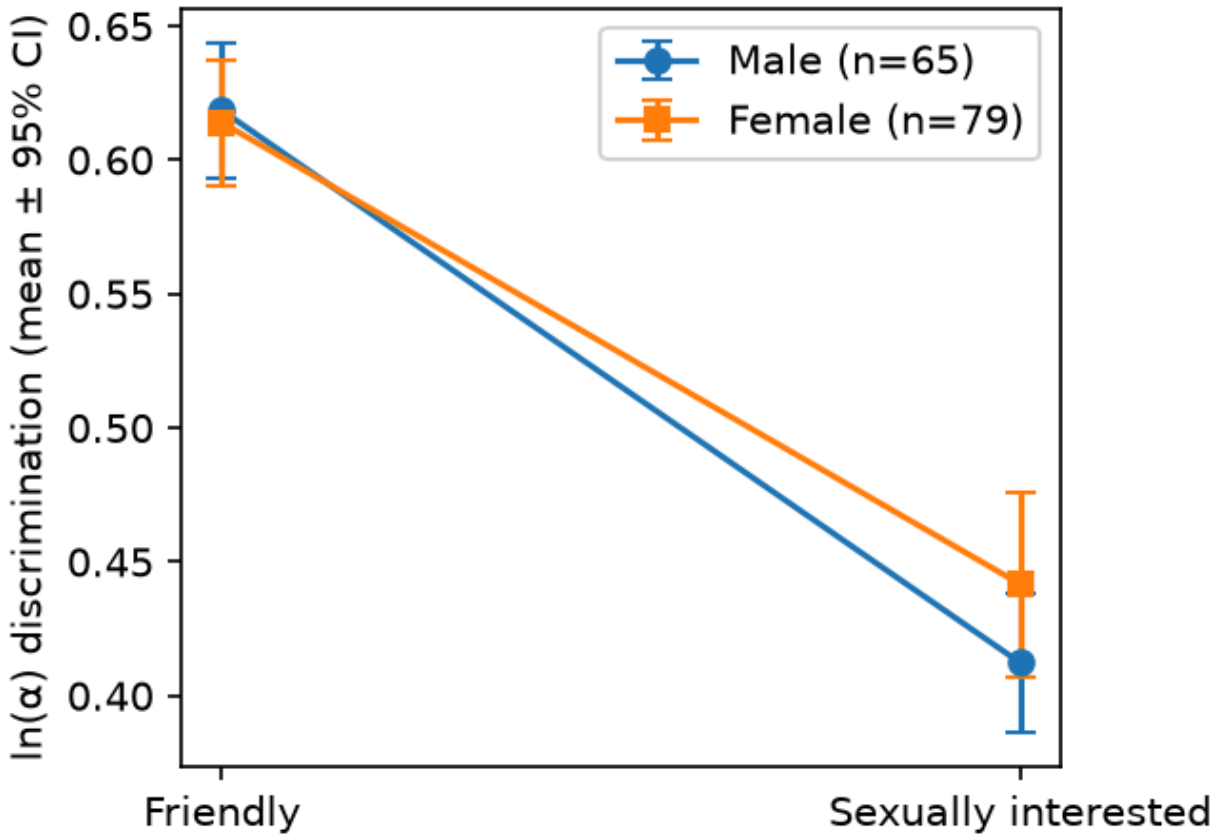
Primary test (fixed before analysis). 2 (category, within) × 2 (gender, between) mixed ANOVA; key term the interaction, equivalent to a one-way ANOVA on Sens_Friend − Sens_SI. The claim holds if the interaction is significant (α = .05) in the predicted direction.

	Men (n = 65)	**Women (n = 79)**
Friendly ln(α), M (SD)	0.619 (0.105)	0.614 (0.107)
Sexually interested ln(α), M (SD)	0.412 (0.108)	0.442 (0.156)
Friendly − SI gap, M (SD)	0.206 (0.156)	0.172 (0.206)

Result. Gender × category interaction: **$F(1, 142) = 1.19$, $p = .276$, partial $\eta^2 = .008$, 95% CI [.000, .061]**. The pattern points the predicted way (men are 0.029 lower than women on sexual interest and 0.005 higher on friendliness, so their friendly–SI gap is 0.034 larger), but the difference is small and not distinguishable from zero: Cohen's d on the gap = 0.18, 95% CI [−0.15, 0.51]; bootstrap 95% CI for the mean difference [−0.025, 0.092].

Main effects: category was large ($F(1, 142) = 147.8$, $p < .001$, partial $\eta^2 = .51$; friendliness was much easier to discriminate than sexual interest for everyone); gender was not ($F(1, 142) = 0.86$, $p = .354$).

Robustness. Welch t on the gap: $t = 1.12$, $p = .264$. Mann–Whitney: $U = 2911.5$, $p = .168$. Adding presentation time (500 vs 3000 ms): gender on the gap $F(1, 140) = 1.00$, $p = .319$; gender × presentation time $F(1, 140) = 1.11$, $p = .294$. Per-category gender differences (Welch): friendly $p = .80$, sexually interested $p = .19$ ($d = −0.22$); the largest male deficit was for rejecting images ($d = −0.32$, $p = .057$), a category outside the claim.



Conclusion. The claim does not hold on this data: the gender × category interaction is small and not significant ($F(1, 142) = 1.19$, $p = .28$), although its direction matches the prediction.

Limitations: with n = 144 the test has 78% power for partial $\eta^2 = .05$ and 40% for .02, so a small effect cannot be ruled out (the upper CI bound is .061).